The most commonly-used evaluation parameters are means.[4] One mean receives the most attention: *the mean effect of treatment on the treated.* This parameter is

$$E(Y_1 - Y_0|X, D=1) = E(\Delta|X, D=1)$$
$$= g_1(X) - g_0(X) + E(U_1 - U_0|X, D=1). \tag{2}$$

The matching methods discussed in this paper focus on estimating an averaged version of this parameter

$$M(S) = \frac{\int_S E(\Delta|X, D=1)dF(X|D=1)}{\int_S dF(X|D=1)}, \tag{3}$$

where S is a subset of the support of X given $D=1$.

This mean answers the question "How much did persons participating in the programme benefit compared to what they would have experienced without participating in the programme?" This parameter is the gross gain to participants from the programme. When compared with costs, this parameter is informative on the question of whether or not an existing programme's benefits exceed its costs and whether the programme should be kept or terminated, provided that the potential outcomes in the no treatment state for all persons are good approximations to the no-programme outcome state for both participants and nonparticipants.[5] It is a nonstandard parameter from the vantage point of conventional econometrics because it combines "structure" (the g_0 and g_1 functions) with the means of error terms (U_0 and U_1).[6]

Social experiments recover the conditional distribution of Y_0, $F_0(y_0|D=1, X)$, if randomization is administered at a stage of the application and enrollment process at which persons would ordinarily be accepted into programme, if the attrition from the programme is random, and if randomization does not disrupt the programme.[7] The evaluation problem arises because ordinary observational data do not provide sample counterparts for the missing counterfactual Y_0 values for participants ($D=1$). Experiments supplement observational data by providing the information needed to form the sample counterpart of $E(Y_0|D=1, X)$ and hence to construct parameter $M(S)$.

To see how randomization solves the evaluation problem, consider randomization among persons who have applied to and been provisionally accepted into a social programme. Persons in the $D=1$ population are selected to receive programme services by a random device. Let $R=1$ if an eligible provisionally-accepted applicant is randomized into the programme; $R=0$ otherwise. It is assumed that if $R=1$, persons accept admission into the programme and receive services and if $R=0$ they do not obtain programme services.

4. Heckman (1992), Heckman, Smith and Clements (first draft 1993, 1997), and Heckman and Smith (1997) discuss other parameters derived from the distribution of outcomes in the programme impacts.

5. Heckman and Smith (1997) and Heckman (1997) consider this parameter in the context of cost benefit analysis and present precise conditions under which it identifies an economically-interpretable parameter. Heckman and Smith (1997) present cost estimates for the programme evaluated here under different assumptions about the social opportunity cost of funds to finance the programme.

6. Heckman and Robb (1985, 1986) present conditions for identifying this parameter using instrumental variables. Their conditions apply to the general "variable treatment effect case" of equations 1(a) and 1(b). See also Heckman (1997) for the implicit behavioural assumptions invoked in using instrumental variables to estimate parameter (2) when responses to treatment are heterogeneous.

7. Randomization at eligibility generates the same information plus the information required to identify Pr $(D=1|X)$. See Heckman (1996).

Recall that it is not necessary to assume that $E(U_1 \mid X)=0$ or $E(U_0 \mid X)=0$, so X can fail to be exogenous in the conventional sense of that term. The lack of any requirement for exogeneity highlights both the unconventional nature of the parameter of interest and the benefits of randomization in estimating it.

We can write observed outcomes for the entire population as $Y = D[RY_1+(1-R)Y_0]+(1-D)Y_0$, so

$$E(Y \mid X, D=1, R=1) = E(Y_1 \mid X, D=1) = g_1(X) + E(U_1 \mid X, D=1), \tag{4a}$$

$$E(Y \mid X, D=1, R=0) = E(Y_0 \mid X, D=1) = g_0(X) + E(U_0 \mid X, D=1). \tag{4b}$$

Randomized-out controls provide the data that can be used to estimate counterfactual (4b). Conditional mean (4a) can be consistently estimated using ordinary observational data on programme participants.

Subtract (4b) from (4a) to obtain

$$\begin{aligned} &E(Y \mid X, D=1, R=1) - E(Y \mid X, D=1, R=0) \\ &\quad = g_1(X) - g_0(X) + E(U_1 - U_0 \mid X, D=1) = E(\Delta \mid X, D=1). \end{aligned} \tag{5}$$

This parameter can be consistently estimated using sample counterparts to population means.[8] One randomization identifies an entire function $E(\Delta \mid X, D=1)$ over any subset of the support of X given $D=1$.[9]

3. MATCHING AS A SUBSTITUTE FOR EXPERIMENTS

Nonexperimental methods use data on members of a nonexperimental comparison group (for whom $D=0$) to infer counterfactual outcomes for participants. A widely-used method is matching. The conventional method of matching estimates parameter $M(S)$, using non-experimental data by assuming that conditional on X, (Y_1, Y_0) and D are independent

$$(Y_1, Y_0) \perp\!\!\!\perp D \mid X. \tag{A-1}$$

where "$\perp\!\!\!\perp$" denotes independence. See, e.g. Rosenbaum and Rubin (1983). If (A-1) is true, then

$$F(y_0 \mid X, D=1) = F(y_0 \mid X, D=0),$$

so conditional on X non-participant outcomes have the same distribution that participants would have experienced if they had not participated in the programme. As a consequence, if the mean exists,

$$E(Y_0 \mid X, D=1) = E(Y_0 \mid X, D=0) = E(Y_0 \mid X), \tag{6}$$

and the missing counterfactual mean can be constructed from the outcomes of non-participants.

If, in addition, it is assumed that

$$0 < \Pr(D=1 \mid X) < 1, \tag{A-2}$$

for all X, then (2) can be defined for all values of X. (A-1) and (A-2) together are called "strong ignorability" by Rosenbaum and Rubin (1983). Under these conditions,

8. Heckman, Ichimura, Smith and Todd (1996b, c) develop, justify and apply nonparametric methods for estimating this parameter.

9. Heckman (1996) shows how randomization acts as an instrumental variable.

experimental and non-experimental analyses identify the same parameters. It is clear that for the purposes of estimating (2) or $M(S)$, the weaker assumption

$$Y_0 \perp\!\!\!\perp D \mid X, \tag{A-3}$$

is enough to identify those parameters.[10]

In the language of Heckman and Robb (1985), matching assumes that selection is *on observables*. Conditional independence or mean independence are strong assumptions. There may be variables apart from X on which the analyst cannot condition that affect both Y_0 and D.[11] In this case selection is on unobservables, as defined by Heckman and Robb. Assumptions (A-1) or (A-3) impose behavioural assumptions that (Y_1, Y_0) or Y_0 do not determine D conditional on X. This rules out selection into the programme based on unobserved (by the analyst) outcomes. More precisely (A-1) implies that $\Pr(D=1 \mid X, Y_1, Y_0) = \Pr(D=1 \mid X)$ while (A-3) implies that $\Pr(D=1 \mid X, Y_0) = \Pr(D=1 \mid X)$. These assumptions are at odds with those invoked in many economic models of self selection such as the Roy model (see e.g. Heckman and Honoré (1990)) and assume either that agents do not act on potential outcomes in deciding to participate in the programme or else that econometricians have as much information about the programme being studied as the agents making decisions. See Heckman and Smith (1997) for a more extensive discussion of the implicit behavioural assumptions that justify the method of matching.

It is often difficult in practice (i.e. with samples of typical size) to match on high dimensional X. This is the matching version of the curse of dimensionality. Rosenbaum and Rubin derive an important practical result. Let $\Pr(D=1 \mid X) = P(X)$. They demonstrate that (A-1) and (A-2) together imply $(Y_1, Y_0) \perp\!\!\!\perp D \mid P(X)$ and hence $Y_0 \perp\!\!\!\perp D \mid P(X)$. This insight shows that matching can be performed on $P(X)$ alone, reducing a potentially high dimensional matching problem to a one dimensional problem, provided that $P(X)$ is known.

By aligning the distribution of observed characteristics in the $D=0$ population with that in the $D=1$ population, matching mimics one feature of randomized data. Randomization within the $D=1$ population ensures that the distributions of X for participants ($R=1$ and $D=1$) and non-participants ($R=0$ and $D=1$) are the same. But there are other features of randomized data that are not so easily achieved by applying the method of matching to nonexperimental data.

A major limitation of nonexperimental methods compared to experimental methods for estimating $M(S)$ is that they do not guarantee that the support for the comparison group equals the support for programme participants. This condition is obviously satisfied in data generated from an experiment, i.e. Support $(X \mid D=1, R=1) =$ Support $(X \mid D=1, R=0)$. The inability to find comparable comparison group members for programme participants is a major source of bias and $M(S)$ often cannot be identified for all subsets S in the support of X given $D=1$. If the support in an experiment differs from the support common to participant and comparison group members in a nonexperimental study, different parameters are implicitly defined and estimated in the two types of studies. Below

10. The assumption of conditional independence of Y_1 given X is useful if the parameter of interest is the mean impact of treatment on the untreated. In that case, estimates of Y_1 for persons with $D=0$ would be inferred from data on persons with $D=1$, instead of inferring estimates of Y_0 for persons with $D=1$ from data on persons with $D=0$ as in this paper. The mean treatment impact on a randomly assigned person can be obtained as the weighted average of the estimates of mean impact of treatment on the treated and on the untreated, under an assumption like (A-1) on both Y_1 and Y_0.

11. Even if one can condition on X in one sample, there is no guarantee that the same variables are available in other samples.

we present some empirical evidence on the importance of this source of noncomparability bias across experimental and nonexperimental studies of the same programme.

Second, both participants and controls in a randomized experiment are usually administered the same questionnaire so outcomes and personal characteristics are measured the same way for both groups. In contrast, observational studies often combine two separate data sets for participants and non-participants that are collected using different survey instruments and different survey definitions of the same economic concept, such as earnings.

Third, both participants and controls reside in the same local labour market. Matching methods are far more effective in recovering the parameter of interest when the comparison group and treatment group both reside in the same local labour markets. For the main body of nonexperimental data that we analyse, programme applicants and nonapplicants come from the same narrowly-defined geographical areas (cities). Both the levels and dynamics of earnings and employment are affected by the conditions of the local labour market in which persons are located.

Table 1 presents features of the nonexperimental comparison groups used in previous evaluations of major U.S. job training programmes. Rows 1 and 2 reveal that few studies have nonexperimental comparison group members located in the same labour markets as programme participants and many use samples that do not administer the same questionnaire to both participants and comparison group members. LaLonde's comparison groups suffer from both defects. A major conclusion of this paper is that placing comparison group members in the same economic environment and administering them the same questionnaire as participants substantially improves the performance of nonexperimental estimators.

4. EXTENSIONS OF MATCHING

Our companion paper extends the received matching framework of Rosenbaum and Rubin in several ways: (1) by developing an asymptotic distribution theory for kernel-based matching estimators both for the cases when $P(X)$ is known and when it is estimated; (2) by demonstrating how the efficiency of the matching estimator can be improved by exploiting exclusion restrictions in terms of variables that appear in outcome and participation equations; (3) by demonstrating how conventional functional form restrictions invoked in econometrics—like additive separability of outcome equations—might improve the efficiency of estimates obtained from matching and (4) by extending matching to a longitudinal setting. A major conclusion of our analysis is that even if $P(X)$ were known, it might be less efficient to condition on it in constructing matches rather than on the original X. (See Heckman, Ichimura and Todd (1993, revised 1997).) We also demonstrate that the ignorability conditions are overly strong for the estimation of (2), or an averaged version of it. All that is required is a weaker mean independence version formulated in terms of $P(X)$

$$E(Y_0 \mid P(X), D=1) = E(Y_0 \mid P(X), D=0). \tag{A-4}$$

(See also Heckman and Robb (1986).) (A-4) is a consequence of (A-1) and (A-2) but can be maintained as a separate and weaker assumption. The theoretical results justified in our companion paper are derived under this assumption.

It is often both conceptually and empirically fruitful to partition X into two components: $X=(T, Z)$ where the T are variables in the outcome equations and the Z are